

# A 3D Multiphase Continuum Model for Atoms and the Periodic Table

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## Abstract

RealQM is a free-boundary multiphase continuum model of atoms and molecules in three space dimensions: non-overlapping unit electron charge densities, continuous across free interfaces whose position is an unknown, interacting by Coulomb potentials without self-repulsion. The ground state is the steady state of a parabolic relaxation of the Coulomb energy. The only number supplied is the nuclear charge  $Z$  — the atoms are neutral, so the electron count is  $Z$  as well — and no constant is adjusted anywhere. Work scales as  $Z$  times the mesh — one Poisson solve per charge density, and a neutral atom has  $Z$  of them — and so linearly in  $Z$  at fixed resolution. Molecules, condensed phases and nuclei are treated in companion articles.

We here let RealQM derive a *periodic table from scratch*: the ground states of the neutral atoms, nothing fitted. The shell structure comes from the build-up, which recovers Li, Be, B and C with no empirical input and is established to carbon; beyond carbon the filling is supplied rather than derived, which is the paper’s principal limitation. Given the structure, the energy is accurate to one or two per cent across the first three periods and to a few per cent over periods 1–4 as a whole, hydrogen to krypton, with no adjustable quantity anywhere.

Two filled packings appear and no third: the antipodal pair, 2, and the octet,  $8 = 4+4$ , two dual tetrahedra. The closed shells are then 2,  $2 + 8$ ,  $2 + 8 + 8$ ,  $2 + 8 + 8 + (2 + 8 + 8)$ ,  $2 + 8 + 8 + (2 + 8 + 8) + (2 + 8 + 8)$ ,  $2 + 8 + 8 + (2 + 8 + 8) + (2 + 8 + 8) + (8 + 8 + 8 + 8)$ ,  $2 + 8 + 8 + (2 + 8 + 8) + (2 + 8 + 8) + (8 + 8 + 8 + 8) + (8 + 8 + 8 + 8)$ . A pair is added only twice in the whole table, on entering periods 4 and 5; oganesson is three pairs and twenty-eight tetrahedra. The row capacities, assembled from the same two units, are 2, 8, 8, 18, 18, 32, 32, and the closures their partial sums 2, 10, 18, 36, 54, 86, 118. The same two packings organise the nucleus in a companion article, 2 and 8 being also the first two nuclear magic numbers. Computed directly from the shell energies, the Allen configuration energy reproduces the electronegativity ordering across period 2 exactly, with a gradient too steep by a factor of about two.

The energies and configurations are computed in spherical symmetry. Three-dimensional results from the same model are included: the separate electron domains of H, He and H<sub>2</sub> with the free interface between them, and an all-electron build-up in which a charge arriving at a closed Li<sup>+</sup> core wraps around it and closes a shell rather than merging into it, declining a merged state that lies 1.055 Ha lower and stays available throughout. Every atom is solved all-electron, with no core radius and no fitted length. The calculation runs to krypton; what is bounded at argon is the build-up. Ionization and excitation are treated elsewhere.

**Keywords:** atomic structure; real-space methods; multiphase wave functions; free-boundary problems; electron packing; Aufbau principle; electronegativity; imaginary-time propagation.

# 1 Introduction

**RealQM** is a model of atoms and molecules in the form of free-boundary multiphase continuum mechanics in three space dimensions, based on non-overlapping unit electron charge densities meeting with continuity at free interfaces and interacting by Coulomb potentials without self-repulsion and with kernel potentials. The ground state is computed as the steady state of a *parabolic relaxation*, or imaginary time stepping, which decreases the Coulomb energy while maintaining continuity of charge density across the interfaces, their position being an unknown of the same dynamics. What is reached is the equilibrium accessible from the state one starts in, so the minimisation is *path dependent* — ordinarily an invisible distinction, since that equilibrium is the minimum, but the one that decides the shell structure of atoms. The model is parameter-free: its inputs are the nuclear charge  $Z$  — the atoms being neutral, the electron count is  $Z$  as well — and the energy minimised is the Coulomb energy alone, with no constant adjusted anywhere. Computational work scales as  $Z$  times the mesh, since each charge density requires one Poisson solve on a grid in three space dimensions and a neutral atom carries  $Z$  of them; the cost is thus linear in  $Z$  at fixed resolution, which allows the simulation of large molecules and, with successive model reduction, opens to clusters of molecules; RealQM applied to molecules, to condensed phases and to nuclei is presented in companion articles [1].

We here take a different route: atoms in their ground configurations, and the periodic table they compose. The model is taken entire, in its smallest implementation, with nothing fitted and nothing carried over from the conventional account.

In an atom the densities arrange themselves about the kernel in *shells*: an inner region of tightly bound charge, then successive outer layers, each shell holding a definite number of densities and separated from the next by a free interface. This shell structure — how many charges occupy each shell — is *not* a result of energy minimisation *only*. It comes from the packing, electron by electron, as the atom is built: a dynamic Aufbau in which each new charge either continues to fill the outermost shell or, if that shell is complete, starts the next. A filled shell is shut to it: the charge in a complete shell is a more rigid arrangement than that of an arriving electron, and resists its inclusion. Given the structure, the minimisation then settles the interfaces and the energy (Section 5.2).

An atom is spherically symmetric and its shells are concentric. The model therefore admits a reduction to a single radial dimension, and it is in that form that the table is computed here. The reduction is a specialisation of the same Coulomb energy, not a second model; but radial averaging keeps only the number of charges in a shell and discards their angular arrangement, which is where the packing lives, so it is the three-dimensional form that decides a packing question (Section 4).

**Codes.** We consider here a *minimal implementation* — minimal in the code, not in the model, which is carried in full — essentially three lines expressing the update of the charge densities, the update of the potentials and the update of the free boundary, iterated to a joint steady state (Section 3). It is used in two forms: spherically symmetric in one dimension, some 175 lines, covering all  $Z$ ; and fully three-dimensional, some 300 lines, used at small  $Z$  (Section 10). The three-dimensional form is the general one, and the one that extends to molecules. Both are short enough to be read end to end, so the claim that nothing is fitted can be checked rather than taken on trust.

**Objective.** Let RealQM compute a periodic table from scratch: the *neutral ground states*, and for each of them the *total energy*, the *shell structure*, and the *Aufbau* by which that structure forms. The result is compared against the reference atomic energies assembled from specialised

treatments [17, 13, 15, 14, 16, 12, 18, 11], each suited to its own range. It should be said at once what those references are, since it bears on how the comparison can be read: a total electronic energy is *not* an observable, so they are not measurements but composites — measured ionisation energies with the relativistic, QED and nuclear-motion contributions removed by calculation, and, beyond argon, Hartree–Fock values computed outright. They therefore already contain standard quantum mechanics, and agreement with them is not a verdict of one theory against nature (Section 5.4). RealQM does not attempt that kind of accuracy in any case, and the test intended here is of a different kind: one parameter-free model, based on Coulomb physics, applied uniformly to every element and asked for the gross structure of the whole table at once — a structure that does not depend on the reference energies at all. Every atom is solved all-electron, with the bare nuclear term throughout and no fitted length anywhere. What the heavier elements require is not a parameter but resolution: the innermost shell contracts as  $1/Z$ , so the mesh is refined where it must be, and nothing else changes. What is computed throughout is the *ground states*, which is what a periodic table is about. Ionization and excitation are treated elsewhere, as are molecules, condensed phases of thousands of atoms, and nuclei, with larger implementations of the same three updates [1].

**The table in two numbers.** It is worth putting the central claim in front of the reader at once. The model has two filled packings and no third: the *antipodal pair*, 2, and the *octet*,  $8 = 4+4$ , two dual tetrahedra. Every closed-shell element in the table is a whole number of each:

|    | $Z$ | pairs | octets | period | content                           |
|----|-----|-------|--------|--------|-----------------------------------|
| He | 2   | 1     | 0      | 2      | 2                                 |
| Ne | 10  | 1     | 1      | 8      | 2+8                               |
| Ar | 18  | 1     | 2      | 8      | 2+8+8                             |
| Kr | 36  | 2     | 4      | 18     | 2+2+8+8+8+8                       |
| Xe | 54  | 3     | 6      | 18     | 2+2+2+8+8+8+8+8+8                 |
| Rn | 86  | 3     | 10     | 32     | 2+2+2+8+8+8+8+8+8+8+8+8           |
| Og | 118 | 3     | 14     | 32     | 2+2+2+8+8+8+8+8+8+8+8+8+8+8+8+8+8 |

Seven noble gases from two packings, with nothing left over and no third unit required. A pair is added only twice in the whole table, entering periods 4 and 5, so oganesson is three pairs and twenty-eight tetrahedra and nothing else. Sections 5.1 and 5.3 establish the two packings and derive the row capacities from them.

Two things this does *not* say, stated here so the claim is not read wider than it is. It fixes how many pairs and octets each closed shell contains, not where they sit radially: the counting cannot distinguish a pair placed innermost from one placed outermost, and that question is three-dimensional (Section 5.3). And it is a decomposition of the row capacities, not a derivation of them — that each capacity serves two periods is not accounted for here, and is the principal thing left open.

**How this stands to the textbook account.** By the *textbook* or *orbital* account we mean the standard explanation of the periodic table: the one-electron hydrogenic states  $s, p, d, f$ , with capacities  $2(2\ell+1)$ , filled in the Madelung order. It should be said at the outset that this is *not* the Schrödinger equation. The  $N$ -electron Schrödinger equation has no orbitals — its solutions are functions on  $3N$  variables and carry no  $s, p, d, f$  labels — and the orbital scheme is an independent-particle approximation laid over it, with the filling rules laid over that in turn. Nothing in this paper disputes the Schrödinger equation. What is compared here is one account of the table’s structure against another.

The comparison is worth making carefully, because the two accounts are strong and weak in different places and it would be easy to overstate either.

What that account does not derive is the *order*. The Madelung  $(n+\ell)$  filling rule has no accepted derivation from the Schrödinger equation; Löwdin set the problem out in this journal in 1969 [5] and it has been returned to since without being closed [6, 7]. The  $4s$ -before- $3d$  ordering in particular has none, and Hund’s rules are likewise empirical regularities rather than consequences. This matters for the present comparison in a specific way: in the orbital account the *doubling* — each capacity serving two periods — follows from Madelung, and is therefore exactly as empirical as Madelung is. Neither account derives it.

Nor is the grouping itself settled. The  $(n+\ell)$  groups close at 2, 4, 12, 20, 38, 56, 88, 120 — the *alkaline earths* — which is the left-step arrangement; the conventional table’s noble-gas closures require a different placement laid over it, and whether helium belongs above beryllium or above neon has been argued for a century. The membership of group 3 is still disputed. So that picture reproduces the table while leaving open both the filling order and, arguably, which table is being reproduced.

Against that, the differences here are these. The capacities come from packing rather than from angular momentum, and 6, 10 and 14 never appear at any row — the  $p$ ,  $d$  and  $f$  sub-shells have no counterpart in this model, while the pair survives, though as two charges at opposite poles rather than as the  $\ell = 0$  harmonic. The closures come out at the noble gases directly, with no second placement rule. The order is partly recovered rather than fitted, by a build-up that is established to carbon. The doubling is not yet derived, but it has a direction here that it has nowhere else: it is contained in a single ceiling function, and that function is what relates an areal capacity to a linear radius (Section 5.3). Section 6 takes the comparison further.

**Comparison with textbook quantum mechanics.** The two rest on the same ingredients — kinetic energy and Coulomb interaction, and nothing else — and differ in the form taken by the wave function. Textbook quantum mechanics uses a single global wave function over a  $3N$ -dimensional configuration space, and the Schrödinger equation for it is *linear*. RealQM uses a sum of one-electron wave functions with local, non-overlapping supports in three-dimensional physical space, the supports themselves being unknowns, which makes it a *nonlinear* free-boundary problem. The physics put in is the same; the form of the description is not, and the difference is worth stating plainly rather than as a verdict.

Density functional theory is the nearer comparison, since it too works in three dimensions. The difference is that in RealQM the electrons retain their individuality, by occupying separate regions of space, and in density functional theory they do not, the state being a single common density.

Exclusion is realised differently in each — by antisymmetry of a function on configuration space, or by non-overlap of domains in three-space — and neither is shown here to be an approximation to the other. What is compared in this paper is what each produces for the periodic table. The reader should take the wider ontological reading advanced elsewhere [1] as speculative, pending evidence stronger than atomic energetics can supply; nothing here depends on it.

## 2 The multiphase atom model

**Notation.** Three symbols recur and are kept distinct throughout.  $Z$  is the nuclear charge, and since only neutral atoms are computed here it is also the number of electrons and so the number of charge densities.  $N$  is the number of points of the uniform radial mesh ( $N = 400$  through argon,  $N = 800$  beyond it).  $N_s$  is the occupancy of shell  $s$ , the number of unit charges smeared into that shell’s density.

RealQM models Coulomb interaction between non-overlapping unit electron charge densities

and point-like kernels, posed as a multiphase free-boundary problem of continuum mechanics. Each electron is one unit of charge spread as a density over a domain of three-dimensional space; the domains tile space without overlapping; the densities meet with continuity on a free boundary, its position determined by the solution rather than imposed; and the only interaction is the ordinary Coulomb one, between the densities themselves and between each density and the kernels: the kinetic energy of the densities and their Coulomb energy are the whole of the functional. We summarise below only what is needed here; derivations are in [1]. For an atom of nuclear charge  $Z$  the  $N$ -electron wave function  $\Psi \in H^1(\mathbb{R}^3)$  has the form

$$\Psi(\mathbf{x}) = \sum_{i=1}^N \psi_i(\mathbf{x}), \quad \psi_i \in H^1(\Omega_i), \quad \int_{\Omega_i} \psi_i^2 d\mathbf{x} = 1, \quad (1)$$

on a partition of  $\mathbb{R}^3$  into non-overlapping domains  $\Omega_i$  with free boundaries  $\Gamma_i = \partial\Omega_i$ . Hartree atomic units are used throughout. The total energy is the standard Coulomb functional

$$TE(\Psi) = \underbrace{\frac{1}{2} \sum_i \int_{\Omega_i} |\nabla \psi_i|^2}_{\text{kinetic}} + \underbrace{\int_{\mathbb{R}^3} W \Psi^2}_{\text{kernel-electron}} + \underbrace{\sum_i \int_{\Omega_i} \sum_{j \neq i} V_j \psi_i^2}_{\text{electron-electron}}, \quad (2)$$

with kernel potential  $W(\mathbf{x}) = -Z/|\mathbf{x}|$  and per-electron Hartree potentials  $V_j(\mathbf{x}) = \frac{1}{2} \int_{\Omega_j} \psi_j^2(\mathbf{y})/|\mathbf{x} - \mathbf{y}| d\mathbf{y}$ ; self-repulsion is excluded by construction ( $j \neq i$  on disjoint domains).

What holds a density against its kernel is then the first term. For a single electron the last term is empty and (2) reduces to the kinetic energy against the kernel attraction alone,  $-\frac{1}{2}\Delta$  against  $-Z/r$ : that balance, and nothing else, fixes the size of the density, and for  $Z = 1$  it is the hydrogen problem. A functional built instead on one common density  $\rho = \sum_i \psi_i^2$  integrates  $\rho$  against itself, so the same balance carries a repulsion of the electron with its own field and the density comes out too diffuse; removing that term afterwards is what a self-interaction correction is for. Here there is nothing to remove.

Stationarity in  $\psi_i$  at fixed  $\Omega_i$  gives the one-electron eigenproblem

$$\mathcal{H}_i \psi_i \equiv \left( -\frac{1}{2}\Delta + W + 2 \sum_{j \neq i} V_j \right) \psi_i = E_i \psi_i \quad \text{in } \Omega_i, \quad \frac{\partial \psi_i}{\partial n} = 0 \quad \text{on } \Gamma_i, \quad (3)$$

with continuity of  $\psi_i$  across the inter-electron interface and the homogeneous Neumann condition together constituting the **Bernoulli free-boundary condition**. Spatial exclusion plays the role of the Pauli principle: the electrons occupy disjoint territories rather than antisymmetrised orbitals.

For an atom the model reduces to spherical symmetry — a specialisation available because atoms are spherical — with  $\Psi$  a function of  $r$  alone, the domains  $\Omega_i$  concentric shells, the free interfaces  $\Gamma_i$  radii  $R_s$ , and the same-shell self-term of a shell holding  $N_s$  charges reduced by  $(N_s-1)/N_s$  (Section 4); the reduced equations are exhibited by the code of Section 10. Every atom reported here is solved all-electron, the nuclear term the bare  $-Z/r$  throughout, with no core radius, no screened kernel and no fitted length anywhere.

### 3 Algorithm: the three-line solver

The system is solved by three coupled relaxations to a joint steady state — the *three-line solver*. We state them for the general case of Section 2; the spherical homogenisation stated there is the special case in which the domains are concentric shells and each  $\psi_i$  depends on  $r$  alone. At each step, for every domain  $\Omega_i$ :

H: one electron, one kernel

He: two electrons, one +2 kernel

H<sub>2</sub>: two electrons, two kernels



Figure 1: The model in three dimensions: charge density as surface elevation over the plane through the kernels, one colour per electron domain, kernels marked in yellow on the base plane. **H**: a single domain. **He**: two one-electron domains on one +2 kernel, meeting at a free interface through the nucleus. **H<sub>2</sub>**: the same two domains with the kernels drawn apart to the bond length. He and H<sub>2</sub> differ only in where the kernels sit, which is this model's account of what a bond is. The quantity drawn is  $|\psi_i|^2$  on the territory domain  $i$  holds, unowned cells left unpainted, so the interface appears as the ridge between the two surfaces rather than as overlapping clouds. Cartesian grid,  $120^3$  over a box of 8 a.u.,  $\varepsilon = h$ . The figure is of the partition; total energies are in Table 1.

1. **Imaginary time stepping** of the amplitude — parabolic relaxation —  $\partial_\tau \psi_i = -\mathcal{H}_i \psi_i$  with  $\mathcal{H}_i$  from (3), relaxing  $\psi_i$  toward the lowest eigenfunction of  $\mathcal{H}_i$ ;  $\tau$  is an artificial time carrying the relaxation and has no physical meaning, only the steady state being used;
2. a **Poisson solve** for the per-electron potential  $V_i$ , which in spherical symmetry is the shell-theorem integral over the other shells;
3. a **boundary move**: each interface  $\Gamma_i$  is shifted in the direction that *decreases the jump* of  $\Psi$  across it, driving it toward the continuity that  $\Psi \in H^1$  demands — the Bernoulli free-boundary condition.

After each step  $\psi_i$  is renormalised; at convergence the boundaries are stationary and the per-electron energy is the Rayleigh quotient, supplying the per-shell energies. In the spherical case the scheme is discretised on a uniform grid  $r_i = ih$  with an explicit step  $\Delta\tau = \frac{1}{2}h^2$  (first order, cusp-limited; Section 7), and each of the three is a single one-dimensional grid sweep on  $r_i = ih$ , so every atom converges in a fraction of a second and the whole periodic table in about a minute on a laptop.

**The algorithm is a relaxation, not a global minimisation.** This is worth stating at the outset because it governs how the results below should be read. None of the three updates searches the space of configurations; each is a parabolic relaxation — the amplitude down the gradient of  $H_s$ , the potential toward its Poisson source, the interface toward continuity — and what they reach together is a *steady state*, a configuration at which all three forces balance. Such a state is a critical point of the Coulomb functional, and the ground state is one; but the relaxation arrives at whichever equilibrium lies at the end of the path it is on, and there is no step at which a lower configuration elsewhere in the space is offered and refused. Energy minimisation here is therefore *path dependent*: what is computed is the equilibrium reachable from the initial state, not the infimum over all states.

Ordinarily this distinction is invisible, because the reachable equilibrium *is* the minimum. It becomes the whole point in Section 5.2, where a lower configuration exists, is never reached, and is not the physical one either — so that the path, and not the infimum, is what selects the structure of the atom.

## 4 What the spherical homogenisation costs

**What the  $(N_s-1)/N_s$  factor is, and what the homogenisation costs.** The factor is not a correction added to the model: it is forced by the homogenisation — by replacing the  $N_s$  discrete charges of a shell with their spherical average, which is what makes the problem radial and hence one-dimensional. In 3D the  $N_s$  electrons of a shell occupy  $N_s$  separate one-electron domains and the  $i \neq j$  rule counts  $N_s(N_s-1)/2$  pairs. Smearing them into one spherical density of charge  $N_s$  makes the Hartree self-energy count  $N_s^2$  pairs, of which  $N_s$  are spurious self-pairs; multiplying by  $(N_s-1)/N_s$  removes exactly those. Nothing is fitted, and the parameter-free character of the model is untouched.

Smearing does a second thing, however, and this one is a genuine loss. It replaces the actual pair *separations* by their spherical average. The charges of a shell sit as far apart as the shell allows — antipodal for  $N_s = 2$ , trigonal for 3, tetrahedral for 4 — and those arrangements repel less than the angular average. For unit charges at a common radius  $R$  (units of  $1/R$ ):

| $N_s$ | homogenised, $q = N_s$ | $\times(N_s-1)/N_s$ | packed | ratio |
|-------|------------------------|---------------------|--------|-------|
| 2     | 2.000                  | 1.000               | 0.500  | 2.00  |
| 3     | 4.500                  | 3.000               | 1.732  | 1.73  |
| 4     | 8.000                  | 6.000               | 3.674  | 1.63  |

So the homogenised shell has the pair *count* right and the pair *geometry* wrong, overestimating intra-shell repulsion by a factor 1.6–2. Most of that is absorbed by the shell radius, which is free and expands until the energy re-minimises; what survives appears as a systematic under-binding, and it is largest at  $N_s = 2$ , where the ratio is largest. Helium is the clean test — one shell, two electrons, no core, no  $r_c$ , and a reference that is exactly known as a nonrelativistic eigenvalue rather than inferred from measurement — and it under-binds by 2.0% (Table 1), with the sign the table predicts.

The thing discarded is precisely the angular correlation, which is what packing *means*. This is why the homogenised model cannot adjudicate a packing question on energetic grounds — it has no angles with which to tell 4+4 from a single 8-shell — and why the capacities of Section 5.1 are established by the counting argument and the 3D solve rather than by the radial energies.

## 5 Results: total energies, packing, and electronegativity

We report three results. The electron packing (Section 5.1) and the all-electron total energies (Sections 5.4 and 5.5) come from the spherically-homogenised (1D radial) solver applied to every electron, with the bare nuclear charge and no fitted length. The electronegativities (Section 5.7) follow from the same shell energies. The energies cover periods 1–4, hydrogen to krypton,  $Z = 1$ –36, with nothing adjusted anywhere: the only quantity that varies across the range is the mesh, chosen by  $Z$  so that the innermost shell stays resolved —  $N = 400$  through argon,  $N = 800$  from potassium — which is a discretisation and not a parameter of the model. What is bounded at argon is not the calculation but the *build-up*: from scandium the next charge belongs inside a shell already occupied, which an outward build-up cannot do, so the fourth row’s configurations are supplied rather than derived (Section 5.6).

## 5.1 Electron packing: 4+4, not 8

The shell packing of each atom is supplied by the build-up of Section 5.2, not chosen by fitting energies. Two findings are robust across  $Z = 2\text{--}18$ . *First*, comparing shells of capacity 2, 4, 6, 8, the **cap-4** packing matches reference total energies best everywhere: a capacity-8 shell over-binds by +9% to +22% (worsening with  $Z$ ), and a closed octet is realised only as **4+4**, never as a single 8-shell — the geometrically expected result, since four unit charges pack as a stable tetrahedron while eight on one shell do not. Here **4+4** means two *radial* shells of four, which is all a spherically homogenised model can mean by it; whether the two tetrahedra close onto a common sphere is a separate, three-dimensional question, taken up in Section 5.3. *Second*, distributing each period’s electrons *evenly* across its cap-4 shells (balanced filling) rather than greedily reproduces the classic configurations 2+2+1 (B), 2+2+2 (C), 2+3+2 (N), 2+3+3 (O) and tightens the energies (e.g. C: +4.4% greedy  $\rightarrow$  −1.1% balanced).

The organising principle is geometric: each electron is a localised cloud whose *size* is set by its distance from the kernel — inner electrons compact, outer ones diffuse — and the shells are the close-packings (capacities 2 and 4) of these clouds. Different packings of the same atom have different energies, and — importantly — the *unconstrained* energy minimum is *not* the physical configuration: in the 1D solver it favours the over-packed capacity-8 shell, the radial model lacking the angular exclusion that makes 8-on-a-shell unstable (a first signal that the closed shell is intrinsically three-dimensional). The physical packing is fixed instead by the build-up (Aufbau) of the atom, electron by electron, which is the subject of Section 5.2: the constraint that the inner shells are already filled removes the over-packed arrangements from the search, and among what remains the energy selects correctly. The capacities themselves are not fitted either — 2 and 4+4 are the  $n = 1$  and  $n = 2$  cases of the counting argument of Section 5.3.

## 5.2 Aufbau: how the shell structure forms

The shell structure of an atom is not a static fact to be looked up but something that *forms*, electron by electron, as the atom is built. Why the structure cannot instead be obtained by minimising over packings is worth stating precisely, because it turns out to be a structural property of the model rather than a weakness of the search.

**The energy is monotone in the packing.** Comparing candidate packings of the same atom, the energy decreases without interior minimum as the electrons are gathered into fewer, more compact shells. The unconstrained minimum is therefore always the most compact arrangement, in which the core is absorbed into the valence, and it is far from the reference value:

|    | minimum | $E$ (Ha) | err   |    | physical | $E$ (Ha) | err   |
|----|---------|----------|-------|----|----------|----------|-------|
| Li | 3       | −8.499   | 13.7% | Li | 2+1      | −7.547   | 0.92% |
| Be | 4       | −18.893  | 28.8% | Be | 2+1+1    | −14.603  | 0.44% |
| B  | 5       | −35.450  | 43.8% | B  | 2+2+1    | −24.710  | 0.23% |
| C  | 6       | −59.214  | 56.5% | C  | 2+2+2    | −37.420  | 1.12% |

These energies are computed at 20 000 steps on the solver’s default mesh, so they differ in the third decimal from Table 1, which uses the finer settings of the periodic-table scan — lithium appears here as −7.547 and there as −7.522. Neither is wrong: each table makes an internal comparison, one between candidate packings of the same atom and the other between atoms, and the settings are held fixed within each. They should not be read across.

The physical configuration is, in each case, the *least* bound of the candidates and simultaneously the closest to the reference. A variational search over packings is thus not merely unhelpful here; it is anti-correlated with the answer.

**What a build-up supplies.** The failure is specific: it occurs only where the search is free to dissolve an already-formed core. An Aufbau construction never offers that. Electrons are added one at a time; the parent configuration is inherited; and a filled packing admits no more. Imposing only those three conditions — and letting the energy choose among what remains — recovers the physical configuration of Li, Be, B and C with no empirical input, the energy merely breaking ties. Nothing is frozen numerically: every candidate is solved with all shells, radii and amplitudes relaxed.

This is where the path dependence of Section 3 becomes the physics rather than a caveat. The three conditions are not restrictions imposed on a search; they are a description of the *path* the atom takes as it forms. An electron arriving at an atom whose inner shells are already occupied cannot reach them — not because a rule forbids it, but because a filled shell is the more rigid of the two. What decides is *relative* rigidity, and nothing else could: the interface moves only where one domain’s amplitude locally exceeds the other’s, so the comparison is between the two charges and not a property of either alone. The compact charge of a complete shell wins against the diffuse charge arriving at it; the newcomer is turned aside and takes the room outside instead. Figure 4 shows this happening — the core unchanged and centred while the arriving charge is deflected around it. The merged configuration remains a legitimate state of lower energy; it is simply not reachable from where the arriving charge starts. The build-up is thus not a constraint added to repair the minimisation. It *is* the minimisation, carried out along the path the atom actually takes, and the physical configuration is the equilibrium at the end of it.

**The shell radius is an attractor, not a memory of the seed.** One objection to a demonstration of this kind is that the interface may simply have stayed where it was put. The three-dimensional runs seed the partition at a radius  $R_0$  through the core’s initial extent, and that seeding fixes nothing thereafter: the interface is free, and where it settles is an output. The test is to vary  $R_0$  and see whether the outcome is the same, and it is. Seeded at  $R_0 = 1.0$  the shell settles at 2.139 au; seeded at  $R_0 = 1.6$  it settles at 2.153 au — a spread of 0.7% on a quantity nothing in the calculation prescribes. A boundary that merely remembered its seed would have kept the 60% difference it was given.

The same runs give the domain charges, which are the other thing to check: the core holds 2.02 charges at mean radius 0.76 au and the arriving electron 1.07 at mean radius 4.0 au, so the two-domain structure is what the relaxation maintains and not an artefact of how it was started. It holds also when the core is left unconfined, claiming the whole box, and the new charge is introduced as a separate ball placed within that territory: the newcomer still takes a region of its own and is not absorbed.

Against the radial solver the three-dimensional shell sits 12% high — 2.15 au against 1.91 au for the same configuration in Section 5.4. That offset is accounted for and is not evidence of disagreement: these runs use a kernel softened to  $\varepsilon = 2h$ , which leaves the well some 9% too shallow at lithium’s 1s peak, and a shallower well gives a wider shell. What the three-dimensional runs are offered for is the mechanism, not the radius.

For Be the balanced condition decides alone, and it gives 2+1+1: the row’s capacity is 8, two cap-4 shells, and the two charges of the period take one place in each rather than two in one. The energy prefers otherwise, and by a converged margin: on a step ladder flat from 16 000 steps to

64000, 2+2 comes out at  $-14.8934$  against  $-14.6028$  for 2+1+1, lower by 0.291 Ha, or 7.9 eV. That is not a competing verdict but the monotonicity stated above, seen at one element. The three candidates fall in order of compactness — a single 4-shell at  $-18.893$  (28.8% too deep), 2+2 at  $-14.893$  (1.54% too deep), 2+1+1 at  $-14.603$  (0.44% short) — so the lower energy is reached by over-binding *past* the reference, and 2+1+1 is at once the least bound of the three and the closest to it. Beryllium is thus the whole pattern in miniature, and it is the reason the energy is not allowed to choose. Nor do we read the outer two charges as a  $2s^2$  pair: row 2 has capacity  $8 = 4+4$  and no pair (Section 5.3), so they are two of a tetrahedron’s four places, and whether they settle antipodally within it is the ambiguity noted below. For B and C the closure condition is doing the work, since the energy alone would continue to fill the inner shell.

**Where the construction is incomplete.** Two limits should be stated plainly, since neither is apparent from the results above.

*The build-up is exhibited as a dynamics only at its first step.* Figures 3 and 4 add one charge to  $\text{Li}^+$  and follow it, and what they illustrate is one of the two cases: the *opening of a new shell*, the arriving charge finding the inner shell closed and taking up position outside it. The other case — a charge joining a shell already open, which must then rearrange to share it — is not shown. Beryllium is its first instance, 2+1 going to 2+1+1 as the arriving charge takes a place in the second cap-4 shell rather than joining the resident one, and it is the natural next step. Nothing here carries the construction further up the row. What supports Li through C is the comparison of *converged* configurations reported above — the energy selecting correctly among those a build-up would leave available — which is a statement about the outcome, not a simulation of the process. We therefore offer the dynamic Aufbau as something the model makes explorable, opened at the first step, rather than as a mechanism already carried through the table.

*And from nitrogen onward even the static rule is undetermined.* A shell holding two charges may be a closed antipodal pair or a half-filled tetrahedron, and no *energy* comparison in the present model distinguishes them; the build-up then either opens a spurious further shell or over-fills the existing one. We report the rule as established for Li through C and open beyond, rather than extrapolate it.

It is worth recording that a constraint the build-up does not currently use can settle that choice, and it is not an energy criterion. Take potassium and calcium, where the same ambiguity recurs one period up: the nineteenth and twentieth charges are either a closed pair or two of a tetrahedron’s four places. On the second reading the tetrahedron completes at  $Z = 22$ , leaving  $36 - 22 = 14$  charges to the next closure — and  $14 \equiv 6 \pmod{8}$ , which by Section 5.3 cannot be composed from octets and at most one pair. The reading is excluded, and calcium’s pair must be closed. So the parity rule has force the local energies lack: it rules out completions that cannot reach a closed shell. Whether feeding it to the build-up as a filter carries the construction past carbon we have not tested, and it would not derive the capacities — it presupposes them — but it is the obvious next thing to try, and it is the reason we call the rule open rather than refuted. It follows that there is no break at  $Z = 18$  in the Aufbau, which stops far earlier;  $Z = 18$  bounds the *energies*, whose configurations come from the capacities and the balanced rule.

### 5.3 Two numbers, not four: the pair and the octet

This model has *two* primitive packings, and there is no third. They are the *antipodal pair*, 2, and the *tetrahedron*, 4 — the close-packings of two and of four unit charge clouds on a shell under Coulomb repulsion, both measured above and neither imported. Eight is not a third primitive: it is 4+4, two dual tetrahedra on the corners of a cube, a single shell of eight over-binding by 9–22%

(Section 5.1). Every shell in every row of the table is a pair or an octet, repeated; in the atom the tetrahedron never closes a shell on its own, appearing only doubled.

The capacity unit is therefore 8, and the table is at its shortest written in that unit —  $2 \mid 8 \mid 8 \mid 2+8+8 \mid 2+8+8 \mid 8+8+8+8 \mid 8+8+8+8$ , seventeen shells rather than thirty-one. The spherically homogenised calculations cannot use it, however: eight charges smeared onto one sphere sit closer to the kernel than  $4+4$  spread over two radii and over-bind by 9–22% (Section 5.1). So each octet is *realised* in the radial model as two shells of four. That is a cost of the homogenisation (Section 4), not a claim that the octet is two concentric objects.

It should be said at the outset how far this is from the  $s, p, d, f$  picture, because the agreement on row lengths can disguise it. That picture has four capacities, 2, 6, 10, 14, being the spherical-harmonic counts  $2(2\ell + 1)$  for  $\ell = 0, 1, 2, 3$ . *Three of those four never occur here.* There is no 6 and no shell of six; no 10 and no shell of ten; no 14. They are not present as approximations, nor as limits, nor under other names — the packing simply has no such arrangement, and a row of length  $8k+4$  or  $8k+6$ , which is what a sub-shell of six or ten would require, cannot be composed at all. What survives is the number 2, and even that survives as a different object: not the  $\ell = 0$  harmonic but two unit charges at opposite poles, geometry rather than angular momentum.

So the two accounts are not variants of one scheme. They agree on the row lengths and on nothing beneath them, and what lies beneath is what bonding geometry reads.

**Where the capacities come from.** Nothing is imported. The basic filled units are the *antipodal pair*, 2, and the *dual tetrahedra*,  $8 = 4+4$ , each established above and neither imported. Composing rows from those two:

| row | composition                 | length | closes at |    |
|-----|-----------------------------|--------|-----------|----|
| 1   | 2                           | 2      | 2         | He |
| 2   | 8                           | 8      | 10        | Ne |
| 3   | 8                           | 8      | 18        | Ar |
| 4   | $2 + 8 + 8$                 | 18     | 36        | Kr |
| 5   | $2 + 8 + 8$                 | 18     | 54        | Xe |
| 6   | $8 + 8 + 8 + 8$             | 32     | 86        | Rn |
| 7   | $8 + 8 + 8 + 8$             | 32     | 118       | Og |
| 8   | $2 + 8 + 8 + 8 + 8 + 8 + 8$ | 50     | 168       | —  |
| 9   | $2 + 8 + 8 + 8 + 8 + 8 + 8$ | 50     | 218       | —  |

Every row length and every closure of the table decomposes into the two packings and nothing else, with no appeal to an orbital count. The last two columns are of course not independent evidence: the closures are the partial sums of the lengths and the lengths are the first differences of the closures, so reproducing one is reproducing the other, and we count it once. The claim is one of decomposition, and the distinction matters: that a row of length 18 *is* a pair and two octets is established below; that the third row *has* length 8 and the fourth 18 is not derived here. The period lengths are supplied to the configuration generator, not produced by it, and the doubling of each capacity across two periods remains open (Section 6). On this reading 18 is not  $9+9$  but  $2+8+8$  — a pair together with two octets — which is also how the fourth row divides in fact, two elements then sixteen; and 32 is four octets outright.

That this closes is not the triviality that any even number is a multiple of eight plus a remainder. The remainder is constrained: the model possesses a pair and an octet and nothing between them, so a row of length  $8k + 4$  or  $8k + 6$  would need half an octet or three quarters of one, and could not be composed at all. What holds instead is

$$2n^2 \equiv \begin{cases} 2 & (\text{mod } 8), \quad n \text{ odd,} \\ 0 & (\text{mod } 8), \quad n \text{ even,} \end{cases}$$

since  $n = 2m$  gives  $2n^2 = 8m^2$ , while  $n$  odd gives  $n^2 \equiv 1 \pmod{4}$ . The remainder is only ever 0 or 2 — never 4 or 6 — so every capacity, at every  $n$ , is a whole number of octets carrying at most one pair, and the pair appears exactly when  $n$  is odd. The leading 2 of rows 4, 5, 8 and 9 is therefore not an irregularity to be excused but the parity of  $n$ .

It should be said plainly what this does and does not do. Written in the model's units the capacity is  $8\lfloor n/2 \rfloor \lceil n/2 \rceil + 2(n \bmod 2)$ , which is identically  $2n^2$  — a restatement of the shell fill, not a derivation of it. What it buys is provenance: the numbers are reached from the pair and the octet without the hydrogenic count entering, and are *then* found to agree with it. The content lies in the constraint above, that the remainder can only be 0 or 2, and in what follows.

The capacities are better given as a construction than as a formula, and the construction needs no notation at all: *to pass from one shell to the next, add  $\lceil n/2 \rceil$  octets and flip the pair on or off* — on when  $n$  becomes odd, off when it becomes even. Every ingredient of that step is one of the two measured units, and no quantity from outside the model enters at any point. In octets the increments run 1, 1, 2, 2, 3, 3, 4, ... and the pair alternates present/absent, giving

$$2 \rightarrow 8 \rightarrow 18 \rightarrow 32 \rightarrow 50 \rightarrow 72 \rightarrow 98 \rightarrow \dots$$

with first differences 6, 10, 14, 18, 22, 26, 30 — arithmetic, second difference a constant 4. That is why the growth of the table looks even rather than lurching: the octet increment advances by one every *two* shells, and the tapering a reader sees in the sequence of closed shells is exactly  $\lceil n/2 \rceil$ .

The repetition of each capacity across two rows is visible in all of this, and in three equivalent guises: each capacity serves two rows;  $\lfloor n/2 \rfloor$  takes each value twice in the closed form; and  $\lceil n/2 \rceil$  takes each value twice in the step. These are one fact seen three ways, not three facts. It is worth noting how narrow that leaves the open question: the doubling is now contained entirely in a single ceiling function, so what wants explaining is precisely why the octet increment advances once per two shells rather than once per shell.

That is not yet answered, but it is worth recording where the answer would have to come from, because the number two is not free. Suppose each capacity is used  $k$  times. The growth of capacity *per period* is then the  $k$ th root of its growth per shell, and at  $k = 2$  that comes out exactly

$$\sqrt{\frac{\text{cap}(n+1)}{\text{cap}(n)}} = \frac{n+1}{n},$$

since  $\text{cap} \propto n^2$  — the ratios being 2,  $\frac{3}{2}$ ,  $\frac{4}{3}$ ,  $\frac{5}{4}$ , ... against 4, 2.25, 1.78, 1.56, ... per shell. So  $k = 2$  is the value that converts the quadratic growth of the capacity into the *linear* ratio  $(n+1)/n$ . If the capacity is an area, going as  $r^2$  with  $r$  advancing by one unit per period, then two periods are what an areal shell costs, and the doubling is the exponent in  $r^2$  rather than an extra rule. No other  $k$  does this:  $k = 3$  gives 1.59, 1.31, 1.21, smoother still, so a preference for smoothness alone would drive  $k$  upward without limit, and it is the dimensionality that picks out two.

We put this no higher than a direction. It is a statement about exponents, and it becomes physics only when something independent establishes that a period corresponds to a unit increment of radius. But it is a direction the packing account has and a filling rule does not:  $(n+\ell)$  contains no length.

**What closure requires, and what precedes it.** A closed outermost octet is reached only when every shell inside it is itself complete, at 2 or at 8. That is what distinguishes a noble gas from an atom that merely carries eight charges on the outside, and it is why the closures fall where they do rather than at every eighth element. Put the other way round: since the outermost octet

is completed only at a closure, *no atom can carry a complete outermost octet without being a noble gas*. The octet outermost is not one property among several that a noble gas happens to have; it is unavailable to anything else.

The converse does not hold, and the reason is worth noting rather than hiding. Helium’s shell *is* the pair, so it closes at 2 and never has an octet outside. And in periods 4 and 5 the configurations place the cap-2 pair outermost — krypton as  $2+4+4+4+4+4+4+4+2$  — so there the octet is not the outer layer either. That is the radial-ordering question raised below: the pair is placed outermost while, in the table, its two charges fill first, at potassium and calcium. Which of the two the atom does is not settled here.

The build-up makes the approach to closure explicit, and gives the element before each noble gas a geometry. Filling a row’s two tetrahedra evenly, the occupancies run  $1+1$ ,  $2+1$ ,  $2+2$ ,  $3+2$ ,  $3+3$ ,  $4+3$ ,  $4+4$ : the last step before closure is always  $4+3$  — one tetrahedron complete, the other one vertex short. That is fluorine,  $2+4+3$ , and chlorine,  $2+4+4+4+3$  (Table 1). So the halogens are not merely “one electron short” as a count; they are short of *one tetrahedral vertex*, a definite vacant site next to a filled tetrahedron, and an arriving charge completes the octet by occupying it. The noble gas is the same structure with that site filled.

The octet rule is then doing far more work than it is usually given. It is not merely the rule for the second and third rows: it is the unit out of which every later row is built, 18 being a pair and two octets and 32 four octets, rather than capacities of their own.

One caution about that grouping. The homogenised solver returns argon as  $2+4+4+4+4$  — five shells, each owning a disjoint radial interval — with free boundaries at 0.249, 0.875, 2.099 and 4.700 au, so the four tetrahedra sit at successive radii in the ratios 2.65, 2.29, 2.16. They are uniformly nested, not paired: were each octet a cube of eight at one radius, the four would fall into two close pairs, and they do not. The bracketing of them as  $(4+4) + (4+4)$  is therefore an *interpretation* of the counts, not an output of this calculation — and it could not be an output, since a radial model cannot place two tetrahedra at a common radius in the first place. Whether the octet closes as a cube at one radius or as two nested tetrahedra is a three-dimensional question, and it is open here.

Because the octet is built as  $4+4$ , the leading octet of rows 6 and 7 may equally be written as two tetrahedra,  $32 = (4+4) + 8 + 8 + 8$ . The count is unchanged; what is open is whether that first group closes into a cube or remains two tetrahedra, and the arithmetic cannot decide it. This is worth recording because it is precisely the region the orbital account assigns to the *f* sub-shell, so it is where the two decompositions would be distinguishable if anything distinguishes them.

The disagreement with the orbital account is the same at every row length, and it is a disagreement about decomposition rather than about number. Where that account makes the octet  $2+6$ , an *s* pair and three *p* lobes, this one makes it  $4+4$ ; where it makes the fourth row  $2+10+6$ , a pair with five *d* and three *p*, this one makes it  $2+8+8$ , a pair and two octets; where it makes the sixth  $2+14+10+6$ , this one makes it  $8+8+8+8$ . Both give 8, both give 18, both give 32. What differs is what the eight, the eighteen and the thirty-two are *made of*, and that is what bonding geometry reads.

Two regularities are in play here and should not be run together. The capacities go as  $2n^2$  and decompose as above; separately, each capacity serves *two* rows, all but the first. The latter is not derived here, and it is worth stating precisely what is missing, because the usual phrasing — “why does each capacity repeat?” — points at the wrong place. The repetitions need no account; what needs one is the *growth*. Reading the table as a sequence of increments 2, 8, 8, 18, 18, 32, 32, an increment is used twice and then grows, so there are two questions: why after exactly two uses, and why to the next capacity rather than to anything else. The second is the sharper, since a third increment of 18 would carry 54 to 72, and 72 is itself a capacity — so the alternative is not

arithmetically forbidden, merely not what happens.

It should also be said that the leftover pair at odd  $n$  may not be one phenomenon. At  $n = 1$  there are no octets for it to be left over from: the pair is the entire shell, it is the innermost and smallest, and “only two fit at that radius” is at least available as an explanation there. At  $n = 3$  and above the pair sits with octets already present and with radius in plentiful supply, so nothing of that kind can be doing the work. The two cases share a term in the parity term and little else, which may be why no single mechanism has accounted for both.

One further caution about the pair at  $n = 3$ . Its radial position and its filling order disagree. The configuration generator places it outermost, and the table’s own filling order places its two charges first, at potassium and calcium. Both can hold only if a shell that fills early lies outside shells that fill later — which is the  $4s$ -before- $3d$  inversion, and therefore the same interpenetration that defeats the spherical homogenisation from the fourth row on (Section 5.6). The leftover pair is thus not an independent puzzle; it is where the d-block problem first shows in the packing account. Read cumulatively, argon, krypton and xenon are one, two and three copies of the same block,  $18 = 2+8+8$ , so that  $Z = 18, 36, 54$  are 1, 2, 3 times eighteen — a pattern invisible in the orbital decomposition, where the three read  $s^2p^6$ ,  $s^2d^{10}p^6$  and  $s^2d^{10}p^6$ . It should be said plainly that this is an arithmetic accident and not a replication principle. Argon’s *total*,  $2+8+8 = 18$ , happens to equal the *next capacity*,  $2 \cdot 3^2 = 18$ ; the two coincide at argon and nowhere else in the table. Krypton is therefore argon plus the next shell, not argon twice over, and the pattern duly fails at radon, which is 86 rather than  $4 \times 18 = 72$ .

We do not claim the familiar  $2n^2$  as a result of this model, that count being the hydrogenic degeneracy; what is claimed is that whatever  $2n^2$  is, it is always composable from the pair and the octet, and in only one way.

**The same two numbers in the nucleus.** The pair and the tetrahedron are not features of the electron. They are the close-packings of *unit charge clouds on a shell under Coulomb repulsion*, and that argument is indifferent to what the cloud is or which sign it carries. A companion article [10] applies the same model to nuclei, protons and electrons entered as equal-mass clouds differing only in the sign of the charge, and it builds them from the same two units: the alpha particle is one electron pair inside one proton tetrahedron,  $2 \mid 4$ , and  $^{16}\text{O}$  is four such units nested,  $2+2+2+2 \mid 4+4+4+4$ , its protons therefore two octets.

The first two nuclear magic numbers are 2 and 8, and  $^4\text{He}$  and  $^{16}\text{O}$  are the two doubly magic light nuclei — exactly the two closures the pair and the octet supply, and exactly the two that model reaches. Beyond 8 the two sequences part company: the atomic capacities continue 18, 32, 50 while the magic numbers continue 20, 28, 50, 82, 126, and the latter are not composable from a pair and octets at all. So what is shared is the *unit*, not the table — the tetrahedron of four unit charges, appearing doubled as the atomic octet and singly as the alpha. That two systems differing by five orders of magnitude in energy select the same two arrangements is the strongest indication we have that these capacities are geometry rather than spectroscopy. We attach no numerical claim to it here: the nuclear binding energies are provisional, the alpha-to-deuteron ratio standing at about 11 against 12.7 observed, with the heavier configurations not yet mesh-converged. Which of the two is the right reading of the fourth row —  $9+9$  or  $(2+8)+8$  — is not settled here; it is worth recording that they are distinguishable in principle, since only the second is expressible in the packings this model possesses — and that the energy does not distinguish them in practice. Composing krypton as the third row’s structure twice over,  $2+4+4+4+4$  repeated, against the same 36 charges packed as  $2+4+4+4+4 \mid 4+3+3+2+3+3$ , the two differ by 1.6 Ha in 2662, or 0.06%; for zinc the corresponding difference is 0.04%. Both are a hundred times smaller than either

configuration’s distance from the reference, so the comparison selects nothing. That is the same finding as Section 5.2’s, one row further up: the energy is not what fixes the structure, and two arrangements of the same charges can sit within a twentieth of a per cent of each other.

## 5.4 All-electron total energies, periods 1–3

Table 1 reports total energies with the balanced cap-4 packing against near-exact nonrelativistic reference values [15].

**What the reference column is.** This deserves saying plainly, because a table of this shape is naturally read as a comparison against measurement, and it is not one. The *nonrelativistic, stationary-nucleus total electronic energy* of an atom is not an observable. No instrument reads it. What is measured is ionisation energies and spectra; the reference values are then *constructed* from those by removing, by calculation, the relativistic, QED and nuclear-motion contributions that the measurement contains and the calculation does not. Reference [15] is explicit that its energies combine experimental data with *ab initio* results, the correlation contribution being obtained by comparing experiment against relativistic complete-valence-space calculations. Beyond argon the values available to us are Hartree–Fock level, that is computed outright.

So the comparison here is model against a semi-empirical composite, not model against experiment. We are not aware of any table of experiment against theory for atomic total energies, and there cannot straightforwardly be one: the quantity a calculation delivers is not the quantity an instrument reads, and bridging the two requires exactly the theory under test to supply the corrections. This does not make the comparison worthless — the composites are accurate, and agreement to one or two per cent against them is a real constraint — but it sets what the per-cent figures mean. They measure distance from a construction that already contains standard quantum mechanics, so they cannot be read as a direct verdict of one theory against nature. The observables this model can be held to without that circularity are of a different kind: electronegativity ordering (Section 5.7), shell radii, and geometry.

The mean relative error is 1.6% and the maximum 4.5% (Si, with S at 4.3%); 13 of the 17 atoms fall within 2.5%. That is a few per cent from a 1D radial solve with no basis sets, no antisymmetry and no two-electron integrals, and a few per cent is the level at which the table should be read.

We claim no more precision than that, and one entry shows why. Argon’s +0.1% is the closest agreement in the table and it is not robust: on the same domain, refining the radial mesh to  $N = 724$  and  $1024$  — where the  $1s$  spans 17 and 25 cells rather than the 10 it spans at  $N = 400$  — moves argon to  $-511.7$  and  $-511.3$  Ha, some 3% *under-bound*, the two finest values agreeing with each other to 0.08%. Both are step-converged: quadrupling the relaxation at fixed mesh moves the energy by 0.005 Ha. So the near-exact argon entry is an artefact of the production mesh, and nothing here rests on it — in particular we do not read a *sign* off this table, since the sign of argon’s error is itself a function of the grid. What the reduction predicts, and what survives refinement, is under-binding; see Section 4 and Section 7.

## 5.5 The fourth row: all-electron energies to krypton

The mesh is chosen by  $Z$  rather than held fixed. Since the innermost shell contracts as  $r_{1s} \approx 4.3/Z$ , a single  $N$  cannot serve the whole table:  $N = 400$  through argon and  $N = 800$  from potassium keeps that shell spanned by a comparable number of cells at both ends, which is the quantity that matters and not the mesh itself. Table 2 reports the fourth row on that basis — all-electron, bare

Table 1: Total atomic energies (balanced cap-4 packing, spherically-homogenised solver) versus near-exact nonrelativistic references [15]. Bare kernel throughout, with no per-element constant. Uniform mesh  $N = 400$  for every atom here, each on its own fixed domain  $D$ , so the entries are comparable across the row (Section 7). Mean  $|\text{error}| = 1.6\%$ , maximum 4.5% (Si); 13 of 17 within 2.5%. The reference column is not measured data: a total electronic energy is not an observable, and these values are composites of experiment and calculation — see the discussion below the table.

| $Z$ | atom | config    | $TE$ RealQM (Ha) | reference (Ha) | error |
|-----|------|-----------|------------------|----------------|-------|
| 2   | He   | 2         | −2.845           | −2.904         | −2.0% |
| 3   | Li   | 2+1       | −7.522           | −7.478         | +0.6% |
| 4   | Be   | 2+1+1     | −14.580          | −14.667        | −0.6% |
| 5   | B    | 2+2+1     | −24.707          | −24.654        | +0.2% |
| 6   | C    | 2+2+2     | −37.446          | −37.845        | −1.1% |
| 7   | N    | 2+3+2     | −54.737          | −54.589        | +0.3% |
| 8   | O    | 2+3+3     | −75.382          | −75.067        | +0.4% |
| 9   | F    | 2+4+3     | −102.485         | −99.734        | +2.8% |
| 10  | Ne   | 2+4+4     | −131.731         | −128.938       | +2.2% |
| 11  | Na   | 2+4+4+1   | −165.545         | −162.255       | +2.0% |
| 12  | Mg   | 2+4+4+1+1 | −201.704         | −200.053       | +0.8% |
| 13  | Al   | 2+4+4+2+1 | −249.405         | −242.346       | +2.9% |
| 14  | Si   | 2+4+4+2+2 | −302.465         | −289.359       | +4.5% |
| 15  | P    | 2+4+4+3+2 | −347.534         | −341.259       | +1.8% |
| 16  | S    | 2+4+4+3+3 | −415.030         | −398.110       | +4.3% |
| 17  | Cl   | 2+4+4+4+3 | −466.105         | −460.148       | +1.3% |
| 18  | Ar   | 2+4+4+4+4 | −527.994         | −527.540       | +0.1% |

kernel  $-Z/r$ , no core radius, no fitted length, nothing adjusted per element — so it extends the table of the previous section rather than being obtained by a different method.

**The shell list and the capacity decomposition are different statements.** These tables report *occupations* — what the filling rule assigns to one atom, shell by shell, as the solver partitions it. Section 5.3 reports *capacities* — what a closed shell can hold, built from the pair and the octet. The two coincide at closed shells and nowhere else:

$$\text{He} = 2, \quad \text{Ne} = 2+8, \quad \text{Ar} = 2+8+8, \quad \text{Kr} = 2+8+8+18,$$

with  $18 = 2+8+8$  in turn. For every other atom there is no complete octet to display — the step before closure is  $4+3$ , one tetrahedron short of a vertex (Section 5.3) — so the long lists of  $2s$ ,  $3s$  and  $4s$  above are not a competing account of the capacities but the occupations themselves. We do not write the octets as  $(4+4)$  in these tables, and the reason is given below: the solver returns the shells uniformly nested rather than paired, so that bracketing would assert a structure the computation does not show.

Two things about this row are worth more than its mean error. The first is the *sign*. Every one of the eighteen entries is under-bound, with no exception and no trend in  $Z$ , and under-binding is what Section 4 predicts: the reduction has the pair count right and the pair geometry wrong, overestimating intra-shell repulsion, and what survives is a deficit of binding. The third period read the other way at  $N = 400$ , and Section 5.4 shows why that reading does not survive refinement —

Table 2: All-electron total energies of the fourth row,  $Z = 19\text{--}36$ , on the  $N = 800$  mesh (references as in Table 1). Configurations are written relative to the argon core,  $[\text{Ar}] = 2+4+4+4+4$ , so that the fourth row’s own partition is visible; the entries are the computed shell occupations, each shell owning a disjoint radial interval. Mean  $|\text{error}| = 3.3\%$ , maximum 6.7% (Se). **All eighteen entries are under-bound**, which is the direction the spherical reduction predicts. The configurations are supplied from the ionisation order from scandium onward and are *not* produced by the outward build-up (Section 5.6); what the row demonstrates is that the solver handles the nesting and returns the energies once a configuration is given.

| $Z$ |    | configuration             | $E_{\text{RealQM}}$ (Ha) | $E_{\text{ref}}$ (Ha) | error |
|-----|----|---------------------------|--------------------------|-----------------------|-------|
| 19  | K  | $[\text{Ar}]+1$           | −584.495                 | −599.230              | −2.5% |
| 20  | Ca | $[\text{Ar}]+1+1$         | −666.030                 | −676.758              | −1.6% |
| 21  | Sc | $[\text{Ar}]+1+2$         | −732.157                 | −759.740              | −3.6% |
| 22  | Ti | $[\text{Ar}]+2+2$         | −830.679                 | −848.410              | −2.1% |
| 23  | V  | $[\text{Ar}]+3+2$         | −906.831                 | −942.880              | −3.8% |
| 24  | Cr | $[\text{Ar}]+4+2$         | −1022.885                | −1043.360             | −2.0% |
| 25  | Mn | $[\text{Ar}]+3+2+2$       | −1111.161                | −1149.870             | −3.4% |
| 26  | Fe | $[\text{Ar}]+3+3+2$       | −1199.067                | −1262.440             | −5.0% |
| 27  | Co | $[\text{Ar}]+4+3+2$       | −1352.957                | −1381.410             | −2.1% |
| 28  | Ni | $[\text{Ar}]+4+4+2$       | −1451.899                | −1506.870             | −3.6% |
| 29  | Cu | $[\text{Ar}]+3+3+3+2$     | −1559.747                | −1638.960             | −4.8% |
| 30  | Zn | $[\text{Ar}]+4+3+3+2$     | −1755.379                | −1777.850             | −1.3% |
| 31  | Ga | $[\text{Ar}]+4+3+3+2+1$   | −1870.260                | −1923.260             | −2.8% |
| 32  | Ge | $[\text{Ar}]+4+3+3+2+2$   | −1995.506                | −2075.360             | −3.8% |
| 33  | As | $[\text{Ar}]+4+3+3+2+3$   | −2116.039                | −2234.240             | −5.3% |
| 34  | Se | $[\text{Ar}]+4+3+3+2+4$   | −2238.117                | −2399.870             | −6.7% |
| 35  | Br | $[\text{Ar}]+4+3+3+2+3+2$ | −2521.800                | −2572.440             | −2.0% |
| 36  | Kr | $[\text{Ar}]+4+3+3+2+3+3$ | −2663.335                | −2752.050             | −3.2% |

argon changes the sign of its error between  $N = 400$  and  $N = 1024$ . So the fourth row, computed where the  $1s$  is resolved, agrees in sign with helium and with resolved argon, and it is the coarse third-period entries that are the exception.

The second is what the row does *not* establish. Its configurations are taken from the ionisation order from scandium onward, because a build-up that adds outward cannot place a charge inside a shell already occupied, and the fourth row requires exactly that. The energies are therefore computed in configurations supplied from outside the model, and the accuracy here says nothing about whether the model would have chosen them. That boundary is the subject of the next section, and it is a boundary in the *build-up*, not in the calculation.

## 5.6 Boundary of the spherical model: where radial nesting fails

The reach of the parameter-free calculation is set by the shape of the table itself. The first three rows have lengths 2, 8, 8, and the homogenisation covers them exactly: the packings 2, 4+4, 4+4 nest, one shell inside the next, and the radial description is the whole of the arrangement. The fourth row is 18 long. What lengthens it is the entry of the transition elements, and with them a filling that is no longer a simple outward sequence — the added charges take the *inner* position, beneath the pair that completes the row above. That placement, and not the reach of the calculation, is

what bounds the *build-up*: the energies run on past argon, parameter-free, once the mesh is refined.

It is worth being exact about what fails, because it is not the geometry. The arrangement beyond argon remains a nested sequence of radial shells: the  $3d$  charges sit at a smaller radius than the  $4s$  pair outside them, which is an ordinary nesting and one the radial solver represents without difficulty — given that configuration it returns the observed removal order across the series (Section 8). What fails is the *order of filling*. Through  $Z = 20$  each added charge goes outside what is already there, and a build-up that adds outward reaches the right configuration. At  $Z \approx 21$  the charge added belongs to a shell *inside* the one already occupied — shown model-independently by the ionisation order, since the  $4s$  electrons are the first removed and are therefore the outer ones. A build-up that only adds outward has no such move, so it opens a further outer shell instead.

The distinction matters for what may be claimed, and two boundaries must be kept apart. The *energies* run to  $Z = 36$ , computed all-electron with no fitted length; but the configurations they are computed in come from the balanced cap-4 rule together with the capacities of Section 6 through argon, and from the ionisation order beyond it — not from a build-up carried out element by element. The *Aufbau* proper — the configuration arrived at by adding one charge at a time — is demonstrated only at its first step and established only through carbon (Section 5.2). There is accordingly no sharp break at  $Z = 18$  in the build-up;  $Z = 18$  is where the arrangement stops being reachable outward, which is a different statement and a weaker one.

Nor is  $Z = 18$  a computational boundary. On a uniform mesh the innermost shell contracts as  $1/Z$  — measured here as  $r_{1s}Z \approx 4.3$ , constant to  $\pm 9\%$  from carbon to argon — so at fixed  $N$  it is eventually spanned by too few cells and the energies degrade, turning positive around iron. That is resolution and not physics, and refining the mesh removes it: at  $N = 800$  the all-electron total energies run to krypton, mean error 3.3% and every entry under-bound, with no trend in  $Z$  and nothing fitted anywhere (Table 2). The relaxation step is explicit,  $\Delta\tau = \frac{1}{2}h^2$ , so the step count must be scaled as  $h^{-2}$  alongside; a refinement taken without that returns a *less* converged answer and is easily mistaken for a mesh effect.

The reader can see all of this directly. The Atom Simulator runs the whole table in a browser, at either mesh, and needs no installation:

[https://claes542.github.io/RealMolecule/atom\\_simulator.html](https://claes542.github.io/RealMolecule/atom_simulator.html)

We would encourage the reader to open it alongside this section. Setting  $Z$  through the first three rows reproduces the configurations of Section 5.4. Continuing past argon is more instructive still, because *nothing appears to go wrong*: the solver goes on producing neatly nested shells and converging. Scandium comes out as  $2+4+4+4+4+1+1+1$  — having no move that puts a charge inside an existing shell, the build-up opens a further outer shell to hold one, and then another. Physically scandium is  $[\text{Ar}]3d^14s^2$ , one of those three charges lying *inside* the other two.

The distinction worth drawing is between representing that arrangement and reaching it. The model represents it without difficulty: told to place the  $d$  charges inside and the  $s$  pair outside, it returns the observed removal order for the whole series, which is the calculation of Section 8 and the reason we there call the placement imposed rather than derived. What the model cannot do is *arrive* at it. The build-up adds outward, so it never offers the inner position, and it fails silently — converging, reporting nothing amiss, and answering the wrong question accurately. Only the reference energies and the removal order reveal it.

## 5.7 Electronegativity from the shell energies

The per-electron energies make one further quantity immediate. Allen’s configuration energy — the mean binding energy of the valence electrons,

$$\chi = \frac{n_s \varepsilon_s + n_p \varepsilon_p}{n_s + n_p}, \quad (4)$$

written here as the occupancy-weighted mean over the valence shells, since the model has no  $s/p$  distinction — is a state property of the neutral atom. It needs no ion, no anion, and no calibration, and it is the quantity that bond polarity is built from. The values below use the all-electron balanced packings with no  $r_c$  anywhere.

|                       | Li   | Be    | B     | C     | N     | O     | F     | Ne    |
|-----------------------|------|-------|-------|-------|-------|-------|-------|-------|
| $\chi$ this work (eV) | 7.29 | 10.18 | 14.69 | 20.68 | 28.81 | 36.69 | 47.93 | 58.39 |
| $\chi$ Allen (eV)     | 5.39 | 9.32  | 12.13 | 15.05 | 18.13 | 21.36 | 24.80 | 28.31 |
| ratio                 | 1.35 | 1.09  | 1.21  | 1.37  | 1.59  | 1.72  | 1.93  | 2.06  |

The *ordering* is reproduced exactly, element for element, across the period: the model places F above O above N above C, and so on down to Li. The *gradient* is too steep by a factor 2.23 (span 51.1 against 22.9 eV), and the discrepancy is not a constant scale — the ratio climbs monotonically from 1.09 at Be to 2.06 at Ne.

The error is thus concentrated at the electronegative end, and its cause is the same one identified in Section 5.6. Radial nesting splits the valence into an inner shell bound too tightly and an outer shell too loosely: in nitrogen the middle shell comes out at 44.3 eV where the physical  $2s$  lies near 25 eV, and with three of the five valence electrons in it, it dominates the mean. The same defect appears with the opposite sign in the outermost shell. One structural cause — the outer tetrahedron placed at too large a radius — therefore produces both the steep  $\chi$  gradient and the light  $p$ -block limitation, and both are removed only by a solve that keeps the angular arrangement, that is by the three-dimensional form.

The result is accordingly strongest where the model is: at the electropositive end, with Be and B within 9 and 19%. As a predictor of bond polarity it is the differences across a row that matter, and those inherit the same overestimate.

## 6 Discussion: packing versus orbitals

The shells  $s, p, d, f$  that organise the conventional periodic table are the *eigenmodes of the one-electron hydrogen atom* — the angular-momentum eigenstates  $R_{nl}(r) Y_{\ell m}(\theta, \phi)$  of  $-\frac{1}{2}\nabla^2 - 1/r$ , their capacities  $2(2\ell + 1) = 2, 6, 10, 14$  merely counting the spherical harmonics at each  $\ell$  (times spin). This is a single-electron, non-interacting spectrum, exact for hydrogen and for nothing else; the  $\ell$ -degeneracy is the special symmetry of the pure  $1/r$  potential, already lifted by screening in every many-electron atom. The shells enter the periodic table by *imposition*: the orbital approximation places each electron in a hydrogenic mode, fills them by the Aufbau order, and enforces exclusion through the antisymmetry of a Slater determinant.

How fully does this *explain* the periodic table? Its capacities are single-electron quantities: the  $s, p, d, f$  counts  $2(2\ell + 1)$  are the spherical harmonics of the one-electron hydrogen spectrum, and  $2n^2$  their sums. Those labels are carried over to many electrons by convention, and the many-electron structure rests on rules that are *descriptive* — the Aufbau/Madelung  $(n + \ell)$  filling order and Hund’s rules are empirical regularities rather than consequences of the Schrödinger equation,

the derivation of the  $(n+\ell)$  rule being the open problem Löwdin set out [5, 6, 7], and the  $4s$ -before- $3d$  ordering in particular has no first-principles derivation within the orbital picture. The table is *reproduced* there by a layered construction: angular momentum for one electron, plus fitted filling rules for many. RealQM relocates the capacities to geometry: 2 and 4 and the octet  $4+4$  follow from close-packing under Coulomb repulsion, and composing rows from those units gives every row length of the table, 2, 8, 8, 18, 18, 32, 32, and indeed every  $2n^2$ , which is always a whole number of octets plus one pair when  $n$  is odd (Section 5.2). The *order* is then partly recovered rather than fitted — a build-up that inherits the parent configuration and closes a filled packing selects the physical configuration of Li through C with no empirical input, the energy only breaking ties (Section 5.2) — though it is undetermined from nitrogen onward, and the doubling of each capacity across two periods is not yet derived — for which, however, see the ceiling function of Section 5.3, which has no counterpart in a filling rule.

A concrete difference is worth noting, because the two pictures decompose the *same* number differently. The octet is 8 in both, but the orbital picture splits it as  $2+6$  (an  $s$  pair and a  $p$  sextet, by angular momentum) while the packing splits it as  $4+4$  (two tetrahedra, by geometry). These are not notational variants, and it is worth being exact about which finding settles it. That a single 8-shell over-binds by 9–22% shows only that the octet must be *two* radial shells, not one; it does not discriminate  $4+4$  from  $2+6$ , which is also two shells. What discriminates them is that among capacities 2, 4, 6, 8 the cap-4 packing matches reference energies best everywhere (Section 5.1). On that evidence the model asserts the geometric division. The first shell is special in both readings for the same reason — it admits no subdivision. In the orbital picture  $n=1$  carries only  $\ell=0$ , so there is no second subshell to interleave with; in the packing picture  $n=1$  is the minimal tiling, one cell per half-shell, the antipodal pair. That is consistent with its being the one capacity in the table that is not repeated.

RealQM proceeds differently. There are no eigenmodes: the electrons are localised, non-overlapping charge clouds, and the ground state minimises the Coulomb energy under spatial exclusion. What emerges is geometric *packing* — the close-packing of mutually repelling charges on a shell (the Thomson problem), whose small-number optima are the antipodal pair (cap-2) and the tetrahedron (cap-4), the octet realised as  $4+4$ . The two ways of realising exclusion, set out in Section 1, have different consequences for shape: the angular eigenmodes — the  $p$  dumbbells, the  $d$  cloverleaves — follow from antisymmetry, the tetrahedra from non-overlap. We make no claim as to which account is the more fundamental. What can be said is that the second admits a directly physical reading: charges that cannot occupy the same place, arranging themselves as far apart as the available room allows.

Spin illustrates the same point, and it is worth being explicit about what replaces it. The capacity 2 is the oldest of the quantities at issue: Pauli introduced it at the end of 1924 as a property of the electron he was careful to call *a two-valuedness not describable classically* [2, 4], declining a mechanical picture of it — he dismissed Kronig’s rotating electron and was brought to the spin interpretation only by Thomas. Two decades later, in his Nobel lecture, he still wrote that he had been “unable to give a logical reason for the exclusion principle or to deduce it from more general assumptions,” and that “I had always the feeling, and I still have it today, that this is a deficiency” [3].

In the present model that particular two-valuedness is not a property of the electron at all. The first capacity is 2 because two unit charges on a shell close-pack as an *antipodal pair*, and non-overlap forbids them the same region, so a pair is spatially divided of necessity; helium is that pair, and the 2-split is where the doubling enters. No two-valued attribute is required to obtain it, and none is present: the electron count comes out the same by a different route. Spin has no role to play here. Spin plays no role in the present model, which is a real limitation and not merely an

omission: singlet–triplet splitting, Hund’s rules, and spin magnetic moments lie outside it entirely (Section 8). We note, without demonstrating it, that the *spatial* content of Hund’s first rule — that electrons keep apart — is enforced here unconditionally by non-overlap, whereas its energetic content is absent.

A regularity of the packing, borne out by the table, sharpens the point: **every period closes as 4+4, never as a single 4**. The two are distinct objects. A single tetrahedron has  $T_d$  symmetry and is *directional* — four vertices, no inversion centre — whereas 4+4 places the second tetrahedron as the *dual* of the first, the pair forming the cube ( $O_h$  symmetry, the eight vertices of the *stella octangula*). A closed, inert shell must be spherically symmetric, and only the cube achieves this; the single tetrahedron is its directional, *reactive* half. The octet rule is then geometric — an atom closes its directional 4 into the spherical 4+4 — and carbon’s tetravalence is precisely the four bond directions of its single tetrahedron, completed to the cube by sharing one electron with each of four partners. (Equivalently, 8 is two *dual* tetrahedra forming one cubic shell, of which the radial 4+4 is the 1D shadow — which is why the radial model places the outer tetrahedron too far out, the limitation of Section 5.6.)

RealQM moreover gives its *own* rationale for the capacities, with no appeal to the harmonic count. Each shell is two *dual half-shells* — the pairing realised geometrically, the two interpenetrating tetrahedra of the octet being the  $n = 2$  instance — and for the octet each half is 1+3, an apex over a triangle. Whether that subdivision continues as  $n^2$  for larger shells, and so returns  $2n^2$ , is the open question of Section 5.2; the packings established here are the pair and the octet. The ring counts  $2\ell + 1$  are the geometric counterpart of the angular degeneracy: for  $n = 2$  a half is 1+3 — an apex over a triangle, i.e. a tetrahedron — and the two duals form the cube. Packing alone thus yields both the magic numbers and their  $2\ell + 1$  substructure, with no angular-momentum operator.

This also explains why 8 is privileged. Only  $n = 2$  gives a half (1+3 = 4) whose double is a *Platonic* closed surface (the cube); for  $n \geq 3$  the half (9, 16, ...) is a regular division with no unique polyhedron. So 8 is a closed, spherically-symmetric *surface*, whereas 18 and 32 are *nested composites* — a cube wrapping interior *d*- and *f*-packings. Three facts follow at once: every period closes as the octet 4+4 (the only available closed surface), the *d*- and *f*-blocks are *buried* (no surface carries them), and the long periods are octet surfaces over interior shells, not larger noble-gas surfaces.

The two pictures agree on the magic numbers 2, 8, 18, 32 because both count arrangements on nested spheres in three dimensions — the same geometry read spectrally (the harmonic count  $2n^2$ ) or geometrically (the packing capacity) — but they decompose them differently: 2+6+10+14 into sub-shells versus 4+4, 4+4+2, 4+4+4+2 into tetrahedra and pairs. A closed shell is spherically symmetric either way; the pictures part company in the angular correlation of open shells and, decisively, in *bonding geometry*. There the geometric picture is the one chemistry already uses: VSEPR and  $sp^3$  hybridisation are geometric repairs grafted onto the orbital model precisely because the bare eigenmodes do not give molecular shape — one must *hybridise* *s* and *p* into four tetrahedral lobes to recover what the cap-4 packing yields directly. RealQM does not claim the orbitals are wrong; it locates them as the *spectral shadow of the one-electron problem*, while the structure of the real interacting atom — and of the chemistry built on it — is geometric packing.

## 7 Numerical properties of the solver

**Spatial discretisation.** We measure the total energy under grid refinement at fixed box size, estimating the observed order  $p$  from  $|E_h - E_{h/2}| \sim h^p$  (Table 3). For He the convergence is clean and **first order** ( $p = 0.94$ ): the nuclear cusp — a kink in  $u$  at  $r = 0$  — caps the formally second-order stencil at  $\mathcal{O}(h)$ , which is also why the light-atom total energies retain a  $\sim 1$ –2% residual at

Table 3: Grid-refinement convergence of the total energy at fixed box (He). The order is first ( $p \approx 1$ ), cusp-limited. Heavier atoms require the grid to resolve the inner  $1s$  ( $r \sim 1/Z$ ): at this coarse  $N$  the Ne and Ar cores are under-resolved and the energies are meaningless, so the production runs use  $N = 400$  (see text).

| Atom | $N$ | $h$ (a.u.) | $E_{\text{total}}$ (Ha) | $ E_N - E_{2N} $ | order $p$ |
|------|-----|------------|-------------------------|------------------|-----------|
| He   | 64  | 0.078      | -2.79543                | —                | —         |
| He   | 128 | 0.039      | -2.82554                | 0.03012          | —         |
| He   | 256 | 0.020      | -2.84121                | 0.01566          | 0.94      |

the production grid.

### Why the production mesh is not refined further, and why refining it would not help.

Every energy reported here is computed at  $N = 400$ , and that is a deliberate ceiling rather than a resource limit. The radial mesh is not the accuracy-limiting approximation: the spherical homogenisation is. Section 4 measures what the reduction costs — it has the pair *count* right and the pair *geometry* wrong, overestimating intra-shell repulsion by a factor of 1.6–2 — and the residual it leaves is of order a per cent, in the under-binding direction. The ladder above puts the discretisation error on the same scale for comparison: successive  $\sqrt{2}$  refinements of helium move the total energy by 0.005 and 0.004 Ha, about 0.2% a step, against the  $\approx 2\%$  the reduction itself accounts for. The model error is an order of magnitude the larger, so refining the radial grid buys convergence *to the reduction’s answer*, which is not the atom’s. A more accurate solution of the wrong equation is not a more accurate atom.

$N = 400$  is also the resolution the three-dimensional solver can reach, at  $400^3$ ; holding the radial runs to the same figure keeps the two halves of the paper comparable and avoids quoting radial numbers at a precision the three-dimensional model could not be checked against. The proper next step in accuracy is angular resolution — restoring the pair geometry the reduction discards — and not a finer  $r$ .

One consequence should be stated plainly, since it cuts against the table. Because the mesh is held fixed rather than converged, individual entries carry a grid uncertainty that can exceed their quoted error: argon moves by about 3%, and changes the sign of its error, between  $N = 400$  and  $N = 1024$  (Section 5.4). Neon is similar in size and non-monotone across the same range. The mean of a few per cent over the table is meaningful; any single entry at better than a per cent is not.

The inner  $1s$  contracts as  $1/Z$ , so a uniform grid must be refined in proportion to resolve it: at the coarse  $N$  above the Ne ( $1s$  at  $r \approx 0.1$ ) and Ar ( $r \approx 0.056$ ) cores are unresolved and the energies diverge. This is why the mesh is chosen by  $Z$  rather than held fixed —  $N = 400$  through argon,  $N = 800$  from potassium (Section 5.5) — so that the  $1s$  is spanned by a comparable number of cells at both ends of the range. It is a resolution requirement and not a boundary of the model: what stops at argon is the outward build-up (Section 5.6), which is a statement about the order of filling and not about the reach of the calculation.

**Where the error is.** It is worth saying plainly which approximation dominates. The kernel is the bare  $-Z/r$ , and near the nucleus a uniform grid does not resolve it: the innermost sampled value is  $Z/h$ , and the  $1s$  it binds spans some eight cells at argon. The consequence is measurable. Argon’s two-electron core comes out 69 Ha too deep against the He-like value, a fifth of it, while

the neutral atom lands within 0.1% — so that agreement is a cancellation of the core error against the rest, not accuracy; and the total moves 10.7 Ha between  $N = 400$  and  $N = 800$ , the relaxation being converged at both. In the light atoms the same effect appears as first-order behaviour, doubling when  $h$  doubles. Beside this the packing, the free boundaries and the Coulomb sums are second-order effects: the unresolved  $1/r$  singularity is the largest single contribution to the errors reported here.

**What convergence means in three dimensions.** The ladder above belongs to the radial solver, where a total energy is quoted and its limit is the question asked. In the present setting — a minimal implementation, on small systems — the three-dimensional calculations display *qualitative* features of the solution rather than precise energies or geometry: which charge holds which region, where an interface stands, whether an arriving charge closes into a shell rather than merging inward. Figures 1 to 4 are of that kind, and are reported as such. What is asked of refinement there is that these features do not change under it, the mesh representing shape rather than serving as a quadrature. The precise numbers in this paper come from the radial solver, where the ladder above applies.

**Iteration and free-boundary convergence.** The imaginary-time residual  $\|\dot{u}_s\|$  and the boundary jumps  $|u_s(R_s) - u_{s+1}(R_s)|$  both decay to steady state; at convergence the jumps vanish, verifying the Bernoulli (continuity) condition at every interface.

**Throughput.** Each step is an  $\mathcal{O}(N)$  one-dimensional grid sweep over the radial mesh; a single atom converges in a fraction of a second and the whole of the first four rows in seconds on a laptop CPU — no basis sets, no two-electron integrals, no  $N$ -body configuration space.

## 8 Limitations

**The shell structure is an input where it ought to be an output.** This is the limitation that matters most, and it bounds what the parameter-free claim is worth. A model handed the configuration and asked only for the energy has had the interesting part supplied: which shells exist and how they fill is what a theory of the atom should *produce*. Here it is produced only at the beginning. The build-up derives it for Li, Be, B and C (Section 5.2); from nitrogen to calcium the balanced cap-4 rule is extrapolated, and that rule was itself adopted because it reproduces the known configurations; from scandium the arrangement is taken from the ionisation order, since the arriving charge belongs inside a shell already occupied and a build-up that adds outward has no move that puts it there (Section 5.6).

The distinction to keep is between capacities and filling. The *capacities* are derived — that is the content of Section 5.3, and they are built from the pair and the octet without  $2n^2$  entering — but the *filling* is not, beyond carbon. So the energies of Sections 5.4 and 5.5 are computed in configurations the model did not choose, and their agreement with reference values is evidence about the model’s energetics and not about its account of structure. Closing that gap, rather than refining the energies, is what would make this an account of the periodic table rather than a calculation performed on one. It is the principal open item of the paper, alongside the doubling of each capacity across two periods.

**Spin.** The model contains no spin, and none is introduced anywhere: exclusion is realised geometrically, by non-overlap of domains, and the electron counts that spin supplies in the orbital picture

are here supplied by the two half-shells — the two-valuedness Pauli declined to picture mechanically being, in this model, the antipodal pair (Section 6). This is a limitation of substance, not of presentation: singlet–triplet splitting and exchange energy as such, Hund’s rules and multiplicity, spin magnetic moments, fine structure, Zeeman and ESR responses, and the ortho/para distinction in helium all lie outside the model as formulated. Nothing reported here depends on them — ground-state energies and shell structure do not — but any extension to magnetic properties or to multiplet spectra would require the model to be enlarged, not merely refined.

Only the stationary (converged) solution is used, so the temporal order of the imaginary-time relaxation is immaterial. The relevant limitation is the *spatial* discretisation: it is low order and cusp-limited (first order at the nucleus, Section 7), so the light-atom total energies retain a  $\sim 1\text{--}2\%$  residual that a higher-order stencil or a logarithmic radial mesh would reduce. For the transition metals the  $d$ -shell must be placed *inside* the  $s$ -shell for the arrangement to be consistent at all; that placement is imposed, not derived, and we do not claim the ordering as a result. The interpenetrating  $4s/3d$  structure is not sorted out here. Open-shell and strongly non-spherical configurations test the angular representation and are flagged where relevant; electron affinities are a known difficult case and lie outside this paper.

## 9 Conclusion

A periodic table has been computed from scratch: the ground states of the neutral atoms, every element solved the same way, with nothing fitted.

Total energies come out within one to two percent across the first three periods, and the interface radii out of the minimisation. The result organises into one statement: packing forms the structure, the Aufbau selects which packing forms, and the energy realises it — none of the three suffices alone.

Two things follow from geometry alone and need no parameter: the octet as  $4+4$ , two dual tetrahedra forming a cube, which a single shell of eight over-binds; and, composing rows from that octet and the antipodal pair, every row length 2, 8, 8, 18, 18, 32, 32 with closures at all seven noble gases — 6, 10 and 14 never appearing, so that the account is not a variant of the  $s, p, d, f$  scheme but a different decomposition of the same row lengths; and the same two units carry the first two nuclear magic numbers in the companion article [10]. The electronegativity ordering across period 2 follows from the shell energies directly.

The reach of the calculation is set by the same geometry. The first three rows have a clear structure — shells nesting simply, capacities 2, 8, 8 — and there the spherically homogenised form captures the essence of the table, an atom being spherically symmetric. From the fourth row the shells interpenetrate, the  $3d$  lying inside the  $4s$  that fills after it; that structure we have not sorted out in detail, and the homogenisation fails there. The three-dimensional form of the model carries no such restriction, and it is the form that extends to molecules.

## 10 Code and reproducibility

The three relaxations of Section 3 are the whole algorithm, so the code implementing them is the simplest an atom admits. Two forms are used:

|                                   | scope                    | solver              |
|-----------------------------------|--------------------------|---------------------|
| spherically symmetric (1D radial) | all $Z$                  | $\approx 175$ lines |
| Cartesian (3D)                    | small $Z$ , no reduction | $\approx 300$ lines |

The counts are of the solver proper — the imaginary time stepping, the Poisson solve and the boundary move, with their initialisation. Each source file carries at its head the three updates *quoted in their own code form*, with the line at which each appears, so that a reader can check the three-line claim against the lines that run rather than against a paraphrase of them. We would make a general point of this. A model whose content is three update rules is more honestly presented by exhibiting those rules as executable text than by transcribing them into display equations and asking the reader to trust that the program implements what is written: the code is where the approximations, the boundary handling and the discretisation actually live, and a short program is a stricter specification than a long derivation. Python or JavaScript of this size is, we think, a reasonable medium for computational physics, and the appropriate object of review alongside the text.

The contrast with established practice is worth stating, and not as disparagement. The standard quantum-chemistry and electronic-structure packages run to hundreds of thousands or millions of lines, written by large teams of computational physicists over decades and without AI assistance of any kind, and they are mature and heavily validated. But no one reader holds such a code in view. In practice its physics is taken on the authority of the community that maintains it rather than read, and the distinction between what the model asserts and what the implementation does is not available for inspection. The codes here are short for a different reason, and not because anything has been conceded to get them short. They are short because the model is nothing but the physics: charge densities in space, the Coulomb interaction between them, and the three updates that carry them to equilibrium. No computational apparatus is laid over the physics to make it tractable, and the length of the code follows from that. Nothing is given up in reach — the same three updates run from atoms to molecules to condensed phases — nor in speed, the work scaling as  $Z$  times the mesh and so linearly in the number of electrons. Nor, measured as accuracy per unit of computation, in accuracy: the whole table to one or two per cent in about a minute on a laptop. What the size additionally buys is that the entire content of the calculation can be read by one person in an afternoon — which for a paper claiming that a model reproduces a periodic table with nothing fitted is not a small thing, since the claim is worth no more than a reader’s ability to confirm it.

There is a practical side to this. The codes here were developed with AI assistance from p5.js templates written by the author, and those templates are essentially the three update lines displayed above: the content of the method was authored directly, and what assistance supplied is the scaffolding around it — initialisation, plotting, batch drivers — none of which enters a result. The reasonable worry about AI-assisted work is that the reasoning behind a program becomes unavailable for inspection, that the code turns into the part of a paper nobody reads. A program short enough to be read in full is the answer to that worry rather than an instance of it: what was written is on the page, in the form a referee reads it, and can be checked line by line against the physics it claims to implement. Not all coding should be hidden by AI, and least of all the few lines that carry the content of a model. The enclosing pages add plotting, a user interface and the batch harnesses that produce the tables, and are larger; nothing in them enters the computation. Neither solver uses a basis set, a library eigensolver, or a fitted functional, and both are short enough to be read end to end. That is the point of reporting their size: the claim that no parameter is adjusted anywhere is one a reader can check directly rather than take on trust.

The molecular and condensed-phase work uses larger implementations of the same three relaxations — a labelled-domain solver and a GPU solver, a few thousand lines each, which reach systems of thousands of atoms. Neither is used for any result reported here; both are described with the work in which they are applied [1].

The complete implementation, the validation suite, and an interactive gallery are publicly avail-

able:

- Source code: <https://github.com/Claes542/RealMolecule>
- Interactive gallery: <https://claes542.github.io/RealMolecule/>
- Atom Simulator, the homogenised solver, running in a browser: [https://claes542.github.io/RealMolecule/atom\\_simulator.html](https://claes542.github.io/RealMolecule/atom_simulator.html)

The last of these is the code whose three updates are quoted above, with a live display of the shells, their free boundaries and the energies as the relaxation proceeds. A reader who wants to know what the model does, rather than what we say it does, will learn more in ten minutes there than from this section. Every figure and table is reproducible from the published source on the hardware stated in Section 7.

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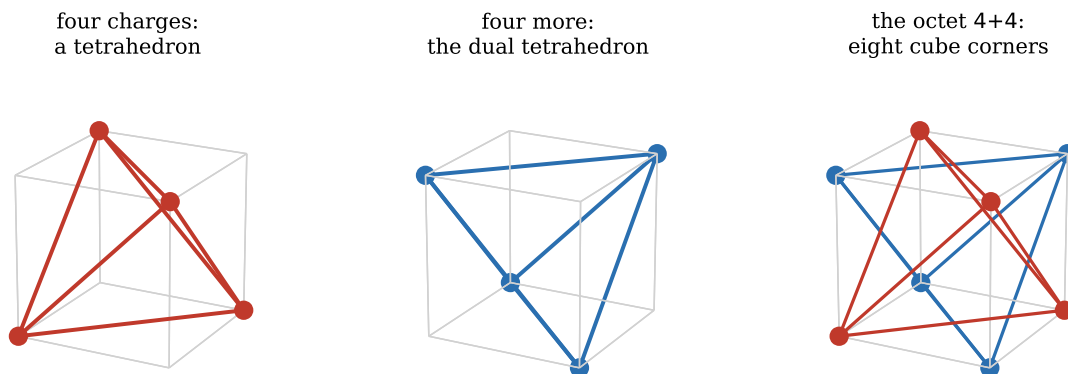


Figure 2: The octet as 4+4. Four unit charges on a shell settle as a tetrahedron, the arrangement that puts them as far apart as a sphere allows. A second four cannot join them — the tetrahedron is closed — but they can occupy the *dual* tetrahedron, and the eight together are the corners of a cube. The two interleave rather than stack: on a cube of side 2 the charges within one tetrahedron are 2.83 apart while the nearest charge of the other is at 2.00, so every charge’s neighbours belong to the opposite four. This is why the closed octet is 4+4 and never a single shell of eight, which over-binds by 9–22% (Section 5.1); and it is the  $n = 2$  case of the counting of Section 5.3, the  $n = 1$  case being the antipodal pair.

The orbital picture makes the same eight as 2+6, an  $s$  pair and three  $p$  lobes. The two accounts agree on the number and differ in the decomposition, and it is the decomposition that bonding geometry sees:  $sp^3$  hybridisation exists to turn 2+6 back into four tetrahedral directions, which 4+4 supplies at the outset.

This is Linnett’s double quartet [8], which reads Lewis’s cubical octet as two interpenetrating tetrahedral sets of four. Linnett separates the two by *spin* — Fermi correlation within a quartet, Coulomb correlation between them. The present model contains no spin, and obtains the same arrangement from Coulomb repulsion and non-overlap alone; what is offered here is therefore not the decomposition but a derivation of it that does not need spin.

The cube drawn here is a *geometric* argument and is not what the calculations of this paper compute. The spherically homogenised solver has no angular coordinate, so its 4+4 is two nested radial shells; measured, the two tetrahedra come out well separated rather than on a common sphere (Section 5.3). The cube is what we expect the three-dimensional form to give, not a result reported here.

One classical result must be conceded, and it bounds the claim usefully. For *four* points on a sphere the Coulomb minimum is the regular tetrahedron (3.6742 against 3.8284 for a square), so the tetrahedron is on firm ground. For *eight* on *one* sphere it is not the cube but the square antiprism, by 0.33% (19.6753 against 19.7408), and this is a theorem [9]. Two things follow. The antiprism is itself a 4+4 — two squares twisted by  $45^\circ$  — so what the theorem refutes is the two-*tetrahedra* reading of a *single* sphere, not the 4+4 decomposition. And this model does not place eight charges on one sphere: a single 8-shell over-binds by 9–22% (Section 5.1) and the octet is realised over two radii, which a single-sphere theorem does not reach. It does mean the cube must not be offered as a Coulomb minimum, and we do not offer it as one.

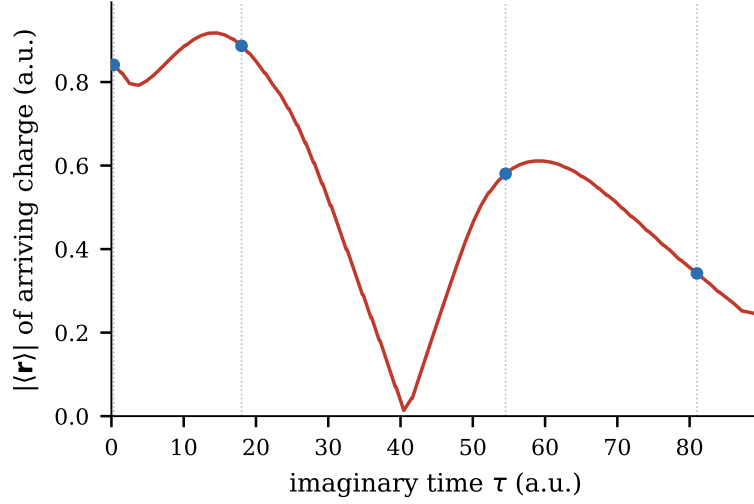


Figure 3: The build-up as a dynamics:  $\text{Li}^+$  with one electron added. The arriving charge is launched as a *half-shell* outside the core — its density concentrated on one side, so that no shell is present at  $\tau = 0$  — and the interface between the two is free thereafter. Plotted is the distance of the arriving charge’s centroid from the kernel against imaginary time. It falls from 0.84 to 0.014 — essentially through the centre — then swings back out: a damped oscillation about the closed shell rather than a decay onto it. The charge wraps *around* the core rather than merging into it. Dotted lines mark the four times shown in Figure 4. A longer run of the same case carries the oscillation through a second pass and down to 4.5% of the launch value; the run plotted here stops partway through the second swing, which is why it ends at 0.24 rather than near zero. Merging is available to the relaxation and is the lower state — a single 3-shell lies 1.055 Ha below 2+1 — so it is declined because the path cannot reach it, not because any constraint forbids it. All-electron, no  $r_c$ , 200<sup>3</sup>; the core is under-bound at this resolution, so no energy or radius is quoted from the run and none is needed for the statement made.

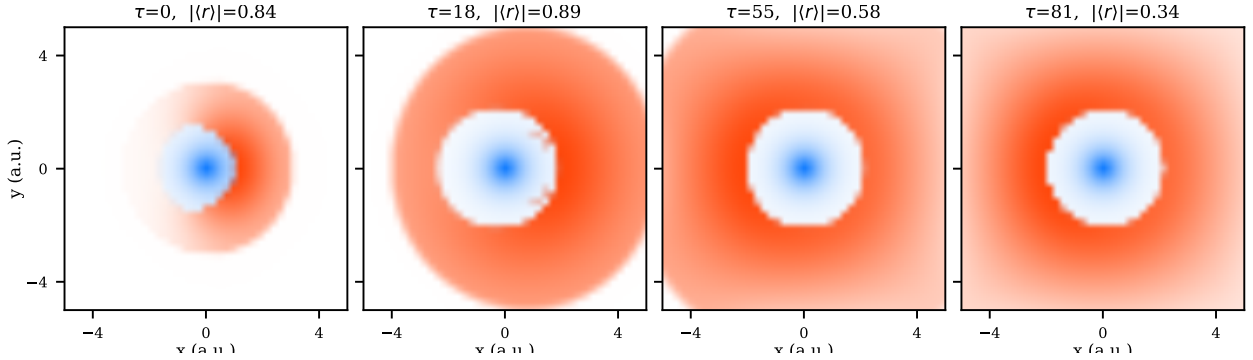


Figure 4: The same run as Figure 3, seen in space: sections through the kernel at four times, with the  $\text{Li}^+$  core in blue and the arriving charge in red, intensity following density on its own territory. At  $\tau = 6$  the arriving charge is a half-shell on one side and there is no shell anywhere; it then sweeps around the core and closes into one. The core is unchanged throughout — centred, compact, holding exactly its two charges at  $r_{50} = 0.66$  a.u. — and at no time does the arriving charge enter it. This is what the non-overlap condition does in practice: the interface moves only where a domain’s amplitude locally wins, and against a formed core it does not, so the only direction available to the incoming charge is around.